Haematological Disorders 1: Haematopoiesis, CBC Interpretation & Iron-Deficiency Anaemia

Internal Medicine Department (lecturer not named on slides) | Source: `1-_Hematopesis_and_CBC_interperetation-Iron_defeciency_anemia.pdf` (49 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The lecture has **no answer key** for its case; the answer below is **[reasoning added; key not on slides]**.

🔗 **Bridge from earlier lectures:**

- **Heart Failure / CAD (Reihan):**
- **anaemic heart failure** = high-output state
- transfusion threshold is higher with **IHD or HF (Hb < 10 g/dL)** than otherwise (Hb < 7)
- *[added]* IV iron improves symptoms in HFrEF with iron deficiency (ferritin < 100, or 100–299 with TSAT < 20 %), the same thresholds as slide 37
- **Tuberculosis (Kotb):**
- TB is the prototype **anaemia of chronic disease** and a cause of **very high ESR (> 100)**
- *[added]* **isoniazid** → **sideroblastic anaemia** (B6 antagonism)
- **Gastric Tumours:**
- **gastrectomy, atrophic gastritis and gastric carcinoma** reduce iron absorption (no HCl → no ferric→ferrous conversion)
- occult GI blood loss → IDA; **an older man with IDA = GI malignancy until proven otherwise** *[added]*
- **FUO & Travel Medicine Pt 1:**
- **malaria** = parasites in RBCs on the blood film
- hookworm (*Ancylostoma*) = travel-acquired IDA
- **Thyroid lectures: hypothyroidism** → non-megaloblastic **macrocytosis**.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Pathophysiology and aetiology of anaemia	Compensations (2,3-DPG, EPO, hyperdynamic circulation)
Knowledge	⭐ Epidemiology, manifestations, complications, management	IDA features (koilonychia, pica); iron dosing; transfusion thresholds
Skill	⭐ Interpret manifestations and investigations	CBC indices; iron panel; microcytic DD table
Skill	Evidence-based management	Oral vs parenteral iron; response timeline
Skill	Recognise emergencies	Transfusion if Hb < 7, or < 10 with IHD/HF

Main subtopics (slide 3):

- blood components and reference ranges (CBC, ESR)
- haematopoiesis sites, stages, requirements
- RBC morphology and inclusions

- **IDA** (pathogenesis, iron panel, treatment)
- **DD of microcytic anaemia** (IDA, ACD, thalassaemia, sideroblastic)

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2. Main Content (Lecture Order)

2.1 Blood Overview & the CBC Report (Slides 4–5)

Plasma:

- straw-coloured, sticky
- **~90 % water**, plus nutrients, hormones, ions (Na^+, K^+)
- **5–7 % protein**: **albumin** (carrier), **globulins** (mainly immunoglobulins), **fibrinogen** (clotting)

Serum = the fluid left **after coagulation** = **plasma minus clotting factors**.

Cells: RBCs, WBCs, platelets.

- **1- RBCs Parameters:**
- Haemoglobin (**Hb**).
- Red blood cell count (**RBCs**).
- Haematocrit (**Hct**).
- Mean corpuscular volume (**MCV**).
- Mean corpuscular Haemoglobin (**MCH**).
- Mean corpuscular Haemoglobin Concentration (**MCHC**).
- Reticulocytic count.
- Red cell distribution width (**RDW**).
- **2- Platelets count (PLT).**
- **3- WBCs parameters:**
- White blood cell count (**WBCs**)
- White blood cell differential (Relative & Absolute Count).

Haematology

CBC-Complete Blood Count

Parameters	Result(s)	Unit(s)	Reference Value(s)
Total WBC	6.55	x10^9/L	4.0-11.0
RBC	6.24	x10^12/L	M:4.5-6.0 F:4.0-5.0
Hemoglobin.	17.2	g/dl	M:14-18 F:12-16
HCT	50.1	%	M:40-54 F:37-47
MCV	80.3	fL	75-96
MCH	27.6	pg	27-32
MCHC	34.3	g/dl	31-35
Platelet Count	210	x10^9/L	150-450
Neutrophils	57.5	%	40 - 75
Lymphocytes	32.3	%	20 - 40
Monocytes	7.8	%	2 - 10
Basophils	0.0	%	0 - 1
Eosinophils	2.4	%	1 - 6
MPV	13.2	fL	6.8 - 10.2

Slide 5: CBC report components with a sample report

CBC section	Parameters
1. RBC parameters	Hb, RBC count, **Hct**, **MCV, MCH, MCHC**, **reticulocyte count**, **RDW**
2. Platelets	PLT count
3. WBC parameters	WBC count; differential (relative and **absolute**)

2.2 Haematopoiesis (Slides 6–10) ⭐

Haematopoiesis = development, differentiation and maturation of **all** blood cells from **pluripotent stem cells**.

- **Erythropoiesis** is regulated by **erythropoietin** (EPO, from the kidney).

Haematopoiesis (Formation of ALL Blood Cells)**:** 'The Process of development, differentiation, and maturation of All Blood Cells'. All blood cells are derived from pluripotent stem cells.

Erythropoiesis is regulated by the Hormone 'Erythropoietin'.

Requirements for Erythropoiesis & Haemoglobin Formation:

- The *Marrow* requires other precursors for effective erythropoiesis: eg.
  - **Metals:**
    - Iron – essential for Haemoglobin synthesis
  - **Vitamins:**
    - Especially Vit. B_{12} - necessary for normal DNA synthesis
    - Folate - necessary for normal DNA synthesis
  - **Amino Acids:**
    - For the production of proteins
  - **Hormones:**
    - ***Erythropoietin***

Slide 6: Requirements for erythropoiesis

Requirements for erythropoiesis (slide 6):

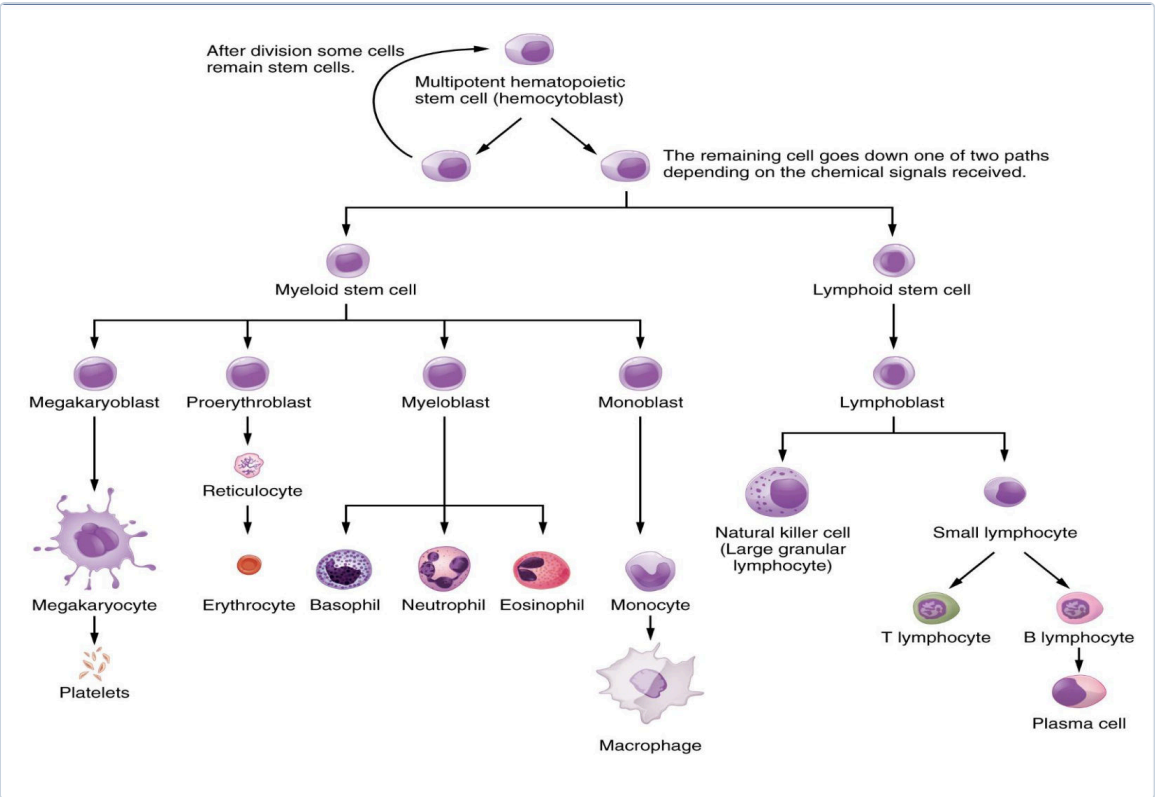
Group	Item	Why
Metals	**Iron**	Hb synthesis
Vitamins	**B12, folate**	DNA synthesis
Amino acids	—	Protein (globin) synthesis
Hormones	**Erythropoietin**	Drive

Sites over a lifetime:

Period	Site
3rd week IU	**Yolk sac**
3rd month	**Liver and spleen**
4th month	Bone marrow starts contributing
Childhood & adult	**Bone marrow only** (medullary cavity)
Adult marrow	**Ends of long bones** (femoral head) + **flat bones** (skull, sternum, pelvis, ribs)
Extreme demand (e.g. chronic haemolysis)	Liver and spleen again = ⭐ **extramedullary haematopoiesis**

Leucocytes:

- **granulocytes and monocytes**: marrow, **myeloid lineage**
- **lymphocytes**:
  - **primary lymphoid organs**: marrow, **thymus**
  - **secondary lymphoid organs**: spleen, lymph nodes, tonsils, adenoids, Peyer's patches



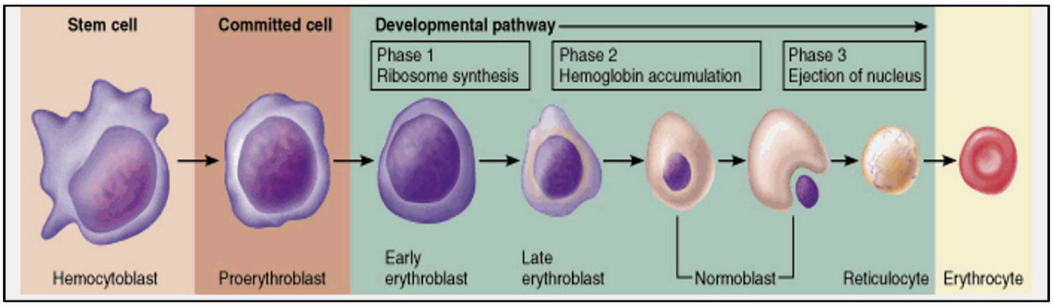
Slide 8: Haematopoietic tree (myeloid vs lymphoid stem cell)

Erythroid maturation: totipotent → pluripotent → committed erythroid stem cell → **proerythroblast → erythroblast → normoblast → reticulocyte** (residual RNA reticulum) → **RBC**.

- **Yolk sac** in the third week of intrauterine life.
- **Liver and spleen** by the 3rd month.
- 4th month of intrauterine life: **BM** starts to contribute
- In adult life, it is restricted to the **ends of long bones** (head of femur) and **Flat bones** (skull, sternum, pelvis, and ribs).

Stages of development

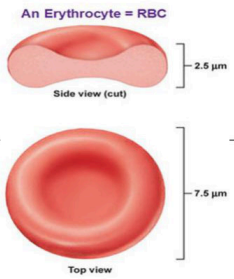
- Totipotent stem cells → pleuripotent stem cells→ stem cells committed for erythropoiesis→ Proerythroblasts→ erythroblasts→ normoblasts→ reticulocytes (containing reticulum of residual RNA) → RBCS



Slide 10: Erythroid maturation (ribosome synthesis → Hb accumulation → nucleus ejected)

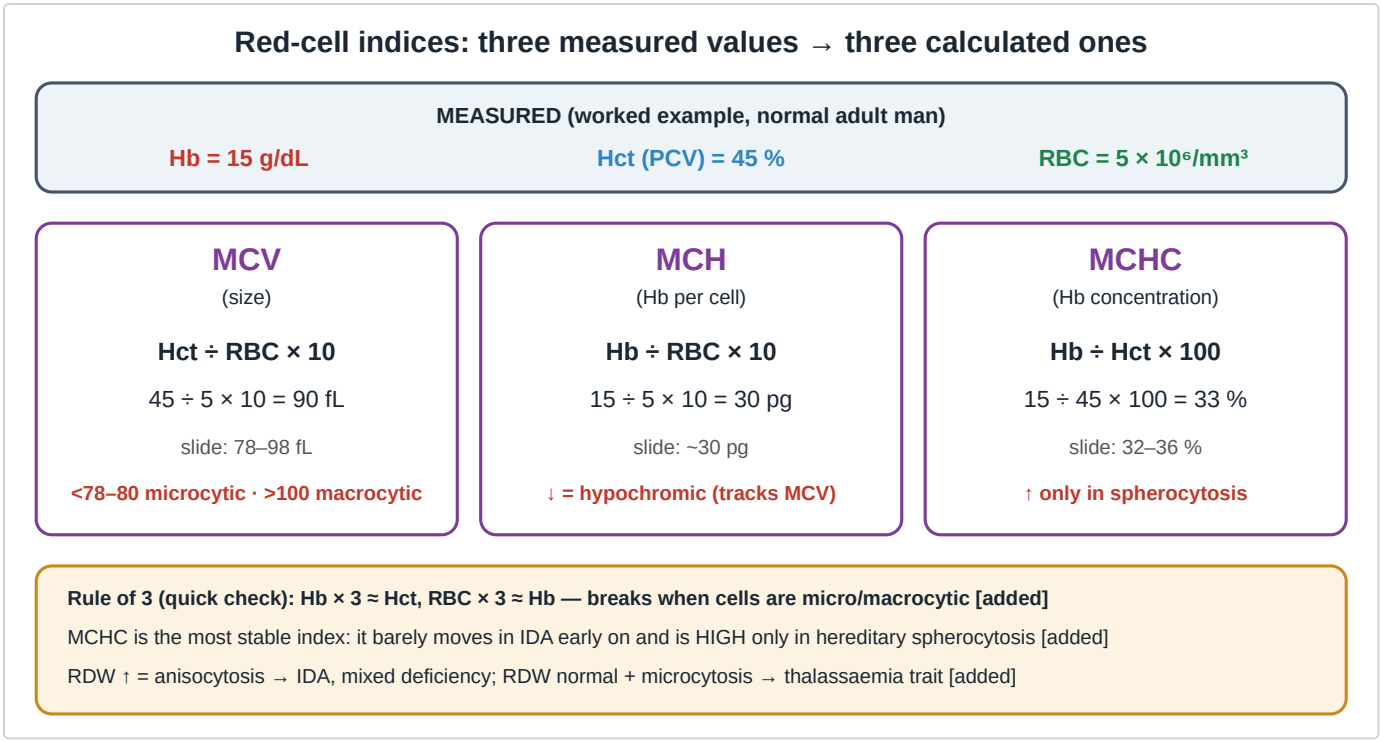
2.3 Haematological Indices (Slides 11–16) ⭐⭐

(1) Shape	(2) Count	(3) Diameter
Normal: Biconcave **Abnormal**: - Spherocytes - Sickle cells - Target cells	**Normal**: ♂: 4.5 – 6 million/mm3 ♀: 3.8-4.8 million/mm3 **Abnormal**: Count = anemia Count = polycythemia	**Normal** = 7.2 micron **Abnormal**: < 6.7 → microcyte > 8 → Macrocyte > 12 → Megalocyte Variable →anisocytosis



Slide 11: RBC shape, count, diameter

#	Index	Normal (slides)	Abnormal / meaning
1	**Shape**	Biconcave	Spherocytes, sickle cells, target cells
2	**RBC count**	♂ **4.5–6** × 10^6/mm^3; ♀ **3.8–4.8**	↓ anaemia; ↑ polycythaemia
3	**Diameter**	**7.2 µm**	**< 6.7 microcyte**; **> 8 macrocyte**; **> 12 megalocyte**; variable = **anisocytosis**
4	**Hct (PCV)**	**~45 %**	↑ **dehydration or polycythaemia**; ↓ **hypervolaemia or anaemia**
5	**MCV**	**78–98 fL**	**> 100 macrocytic**; **< 78 microcytic**
6	**Hb**	♂ **13.5–16** g/dL; ♀ **11.5–13.5**	↓ anaemia or hypervolaemia; ↑ polycythaemia or dehydration
7	**MCHC**	**32–36 %**	= Hb ÷ Hct × 100
8	**MCH**	**~30 pg**	= Hb ÷ RBC × 10
10	**Reticulocytes**	**0.2–2 %**	Index of **marrow activity**



Added diagram: How the red-cell indices are calculated (worked example)

> ⚠️ **Slide inconsistencies:**
>
> - **MCV cut-off:** slide 12 = **< 78 / 78–98 / > 100**. The flowchart (slide 28) = **< 80 / 80–100 / > 100**. Lecture 2 uses **> 115** for megaloblastic. Learn **< 80 microcytic, > 100 macrocytic**; the megaloblastic range is often > 115.
> - **Hb lower limits:** slide 13 gives ♀ **11.5–13.5 and** ♂ **13.5–16**, but slide 22 defines anaemia as **Hb < 12 (♀) / < 13 (♂)** (WHO). Use **< 13** ♂ **/ < 12** ♀ for the definition.

ESR (slides 14–15):

- **Mechanism:** RBCs carry **negative charges** and repel each other. **Albumin keeps the charge; fibrinogen and globulins neutralise it** → **rouleaux** → faster sedimentation.
- **Method: Westergren** (blood + sodium citrate); read at the end of the 1st and 2nd hours.

Normal ESR	1st h	2nd h
Male	**5 mm**	8 mm
Female	**10 mm**	16 mm

Cause of ↑ ESR	Cause of ↓ ESR
○ ↑ Fibrinogen: In pregnancy & menses. ○ ↑ Globulin: ▪ Multiple myeloma. ▪ Collagen diseases. ▪ Malignancy ▪ Infection. ○ ↓ Albumin: Nephrotic syndrome	○ Polycythemia. ○ Afibrinogenemia. ○ Heart failure.

o**Value:** ESR is used to follow up disease progress & response to treatment not for diagnosis

○ **Very high ESR(> 100 mm/HR):**

1. Malignancy
2. Collagen diseases
3. Active TB

○ **Anemia with low ESR**: 1. Sickle cell anaemia 2. Hereditary spherocytosis

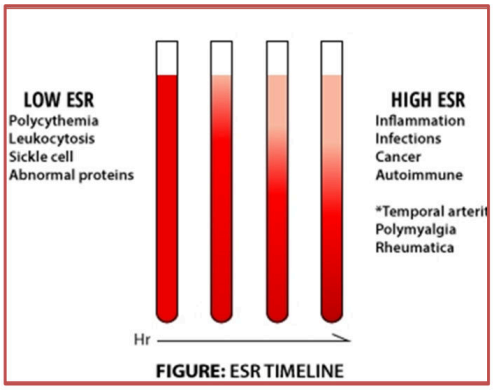


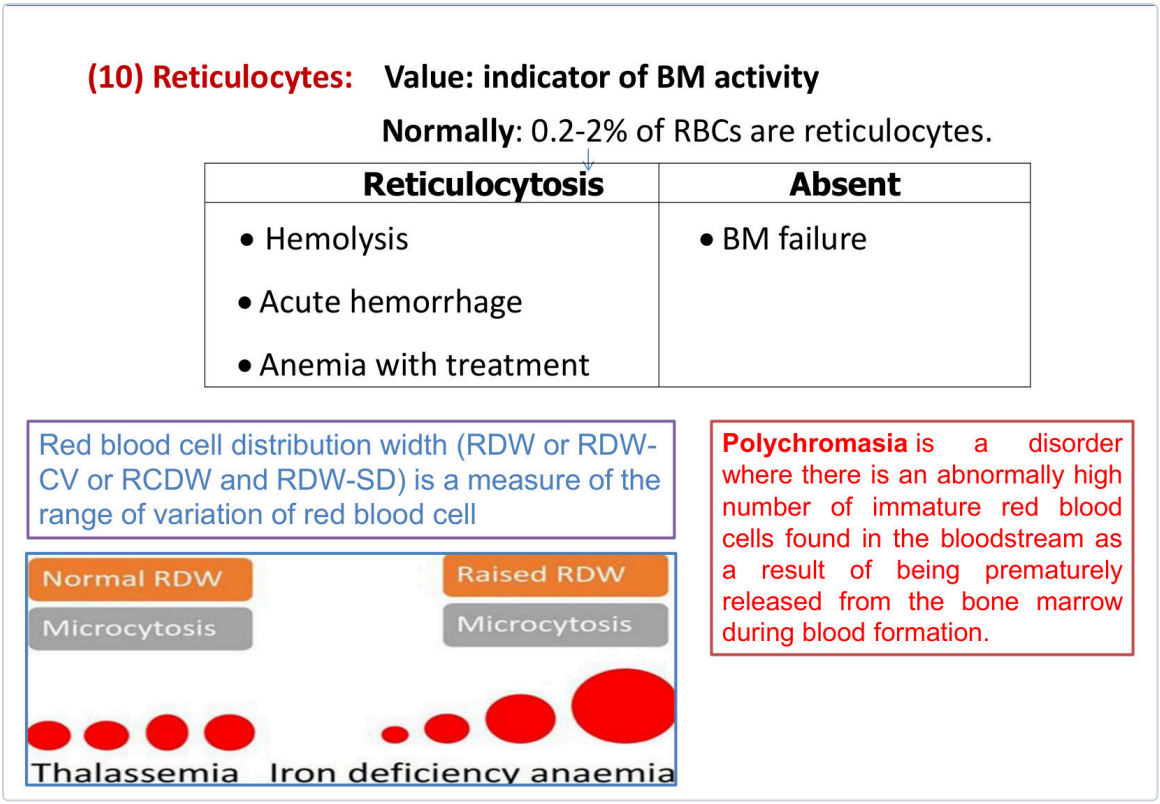
FIGURE: ESR TIMELINE

Slide 15: Causes of high and low ESR

↑ ESR	↓ ESR
↑ **Fibrinogen**: pregnancy, menses	**Polycythaemia**
↑ **Globulin**: **multiple myeloma**, collagen disease, malignancy, infection	**Afibrinogenaemia**
↓ **Albumin**: **nephrotic syndrome**	**Heart failure**
[added] anaemia itself raises ESR	**Sickle cell anaemia, hereditary spherocytosis** (abnormal shape → no rouleaux)

Using the ESR:

- ⭐ ESR is for **following disease progress and treatment response**, **not diagnosis**.
- ⭐ **ESR > 100 mm/h: malignancy** (*[added]* esp. myeloma), **collagen disease** (*[added]* GCA/PMR), **active TB**.



Slide 16: Reticulocytes, RDW, polychromasia

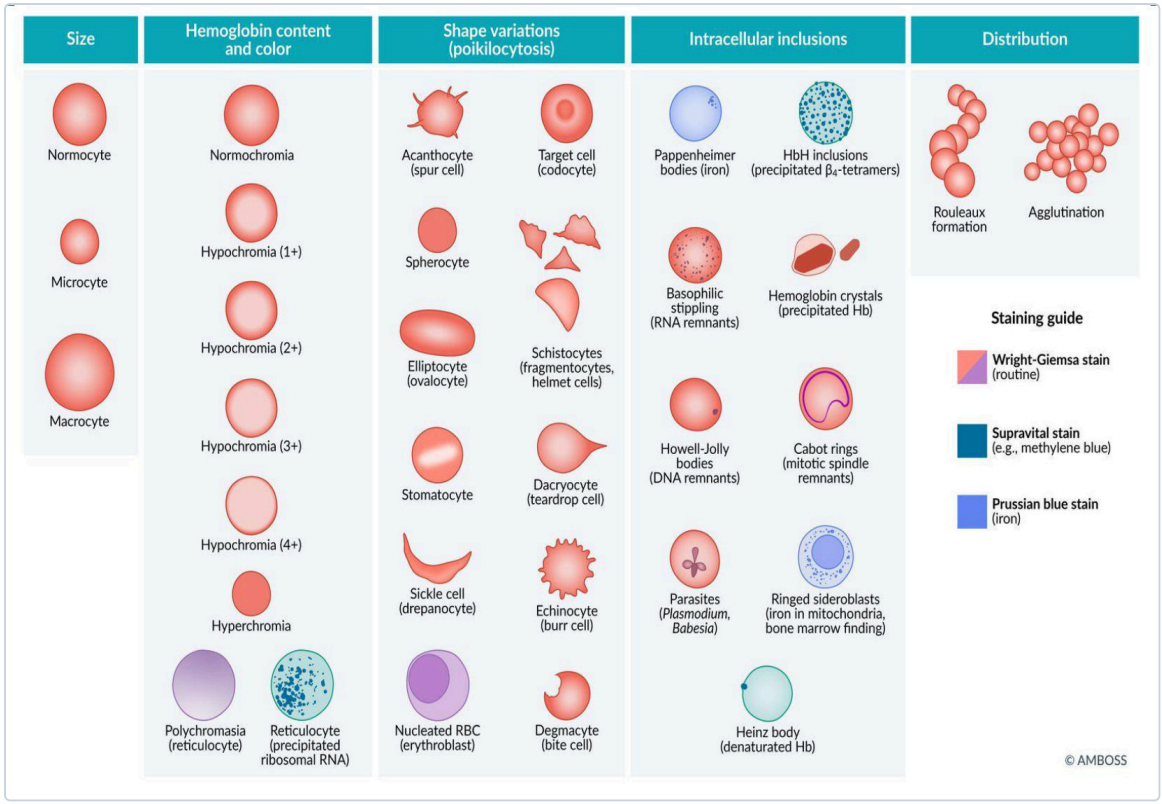
Reticulocytosis	Absent reticulocytes
Haemolysis, **acute haemorrhage**, **anaemia under treatment** (e.g. after iron/B12)	⭐ **Bone-marrow failure**

Related measures:

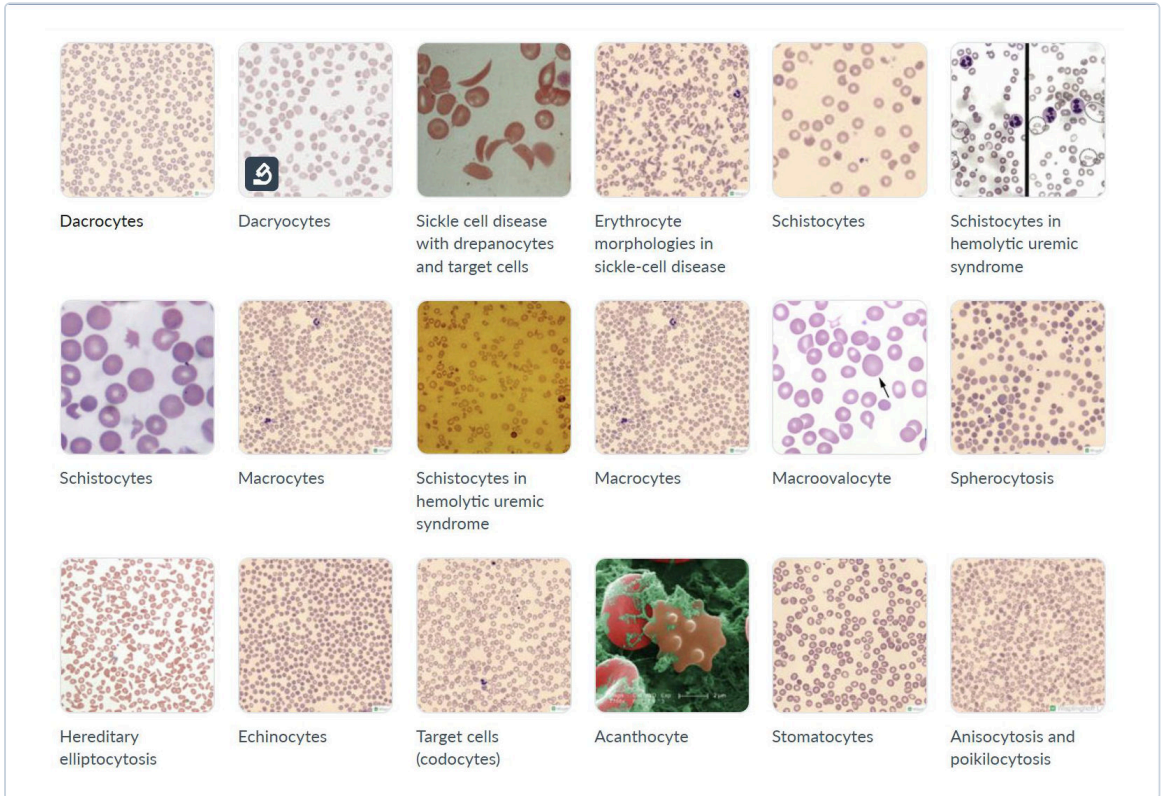
- **RDW** = range of variation in RBC size. ↑ **RDW** = anisocytosis (e.g. **IDA**); **normal RDW + microcytosis** → **thalassaemia** (slide image).
- **Polychromasia** = many immature RBCs (reticulocytes) released prematurely from the marrow → bluish cells on the film.

2.4 Value of the Blood Film (Slides 17–20) ⭐

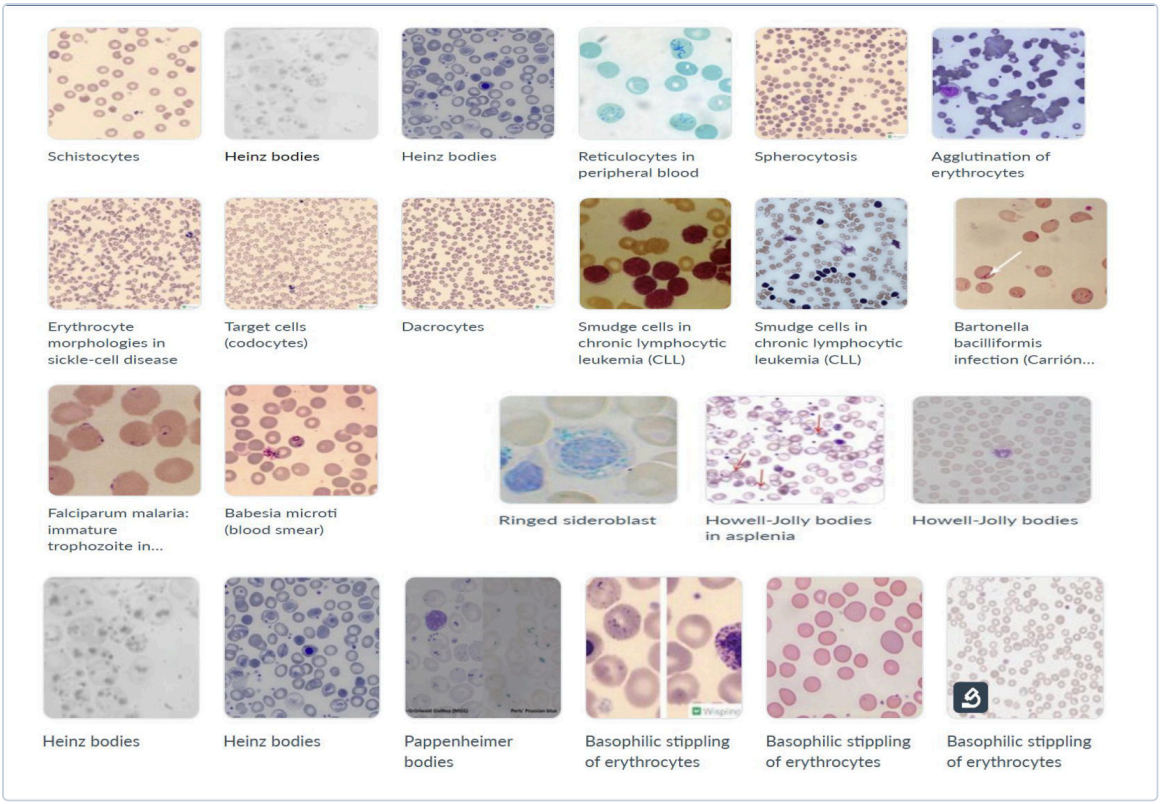
Finding	Meaning
Anisocytosis (size variation)	Megaloblastic or iron deficiency (non-specific); ↑ **RDW**
Poikilocytosis (shape variation)	Non-specific
⭐ **Basophilic stippling** (punctate basophilia)	**Lead poisoning, thalassaemia**
Polychromasia	Causes of ↑ reticulocytes
Intra-RBC parasites	**Malaria**
⭐ **Howell-Jolly bodies** (nuclear remnants)	**Megaloblastic anaemia, post-splenectomy**



Slide 18: RBC morphology atlas (size, Hb content, shape, inclusions)



Slide 19: Film examples: dacrocytes, sickle cells, schistocytes, spherocytes, target cells, acanthocytes



Slide 20: Inclusions: Heinz bodies, Pappenheimer bodies, Howell–Jolly bodies, basophilic stippling, ringed sideroblasts

[added] **Shape-to-disease quick list:**

Cell	Disease
Schistocytes	MAHA (DIC, TTP, HUS, prosthetic valve)
Spherocytes	HS, warm AIHA
Target cells	Thalassaemia, liver disease, HbC, post-splenectomy
Teardrop cells	Myelofibrosis
Bite cells / Heinz bodies	**G6PD deficiency**
Acanthocytes	Liver disease
Pencil cells	IDA

> 🔁 **Recap: CBC basics.**
> - **Formation:** marrow is the only adult site (extramedullary under stress); requirements = iron, B12/folate, amino acids, EPO.
> - **Indices:** MCV < 80 / > 100; MCHC 32–36; retics 0.2–2 % (absent = marrow failure).
> - **ESR:** monitoring, not diagnosis; > 100 = myeloma/collagen/TB.
> - **Film:** stippling = lead/thal; Howell–Jolly = megaloblastic/asplenia.
>
> 🔗 **Link:** the indices exist to **classify anaemia by MCV**, which is the backbone of the rest of the lecture.
>
> ▶️ **Up next: Anaemia in general.**

2.5 Anaemia: Definition & Compensation (Slide 22) ⭐

Definition:

- ↓ O_2-carrying capacity of blood
- **Hb < 12 g/dL (♀)** or **< 13 g/dL (♂)**

Compensatory mechanisms:

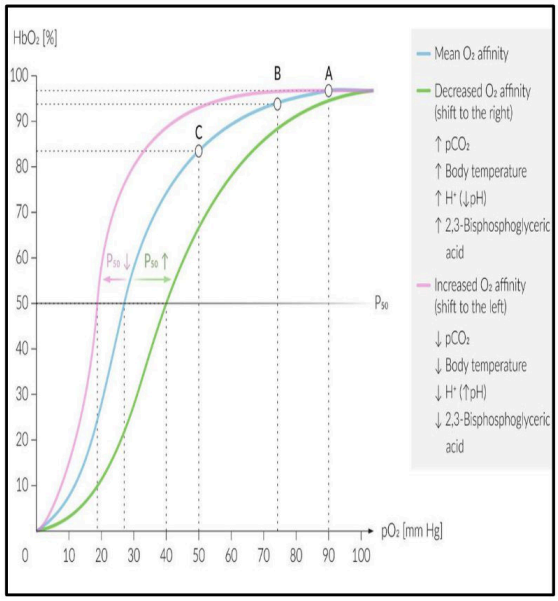
1. **Hyperdynamic circulation**: short circulation time, **high cardiac output**
2. ↑ **Erythropoietin** (kidney) → more RBCs
3. ⭐ ↑ **O_2 delivery**: ↑ **2,3-DPG** → **right shift** of the O_2-dissociation curve (Hb releases O_2 more easily)
4. **Redistribution** of blood to vital organs (brain)

Definition

- Reduced O2 carrying capacity of blood.
- It's identified as Hb < 12 gm% in Females or Hb < 13 gm% in males.

Pathophysiology:

1. Hyperdynamic circulation → Short circulation time & high cardiac output.
2. ↑ Erythropoietin by the kidney→ +++ RBCs
3. ↑O2 delivery to the tissue (shift of O2 dissociation curve to the right): This is due to excess production of 2,3-DPG (2,3-diphosphoglycerate), which makes the binding between O2 & Hb very weak.
4. Redistribution of the blood: blood is shifted to the vital organs e.g., the brain.



Slide 22: O_2-dissociation curve (right shift with ↑ 2,3-DPG)

2.6 Anaemia: Clinical Picture (Slides 23–24)

A. General features of any anaemia:

System	Features
Pallor	Conjunctiva, palms, nail beds
CNS	Dizziness, **tinnitus**, poor concentration, blurred vision; tingling/numbness (peripheral neuropathy)
CVS (hyperdynamic)	**Tachycardia, palpitations**, capillary pulsation, **wide pulse pressure (water-hammer pulse)**, **functional (flow) murmurs**, ⭐ **anaemic heart failure**
GIT	Dyspepsia, anorexia
Renal	Polyuria, proteinuria (hypoxia → renal vasodilatation)
Reproductive	Amenorrhoea or menorrhagia
Oedema	Hypoxic capillary leak + **hypoproteinaemia** (co-existing malnutrition) + anaemic HF

B. Features specific to each anaemia. **C.** Features of the cause.

2.7 Anaemia: Investigations, Treatment, Classification (Slides 25–28) ⭐⭐

Investigations:

- **A.** All anaemias: ↓ RBCs and PCV, ↑ ESR

- **B.** Type by indices (below)
- **C.** Cause of the anaemia
- **D.** Underlying aetiology

Type	MCV	MCH	MCHC
Microcytic hypochromic	↓	↓	↓
Normocytic normochromic	N	N	N
Macrocytic normochromic	↑	N	N

Treatment of any anaemia (slide 26):

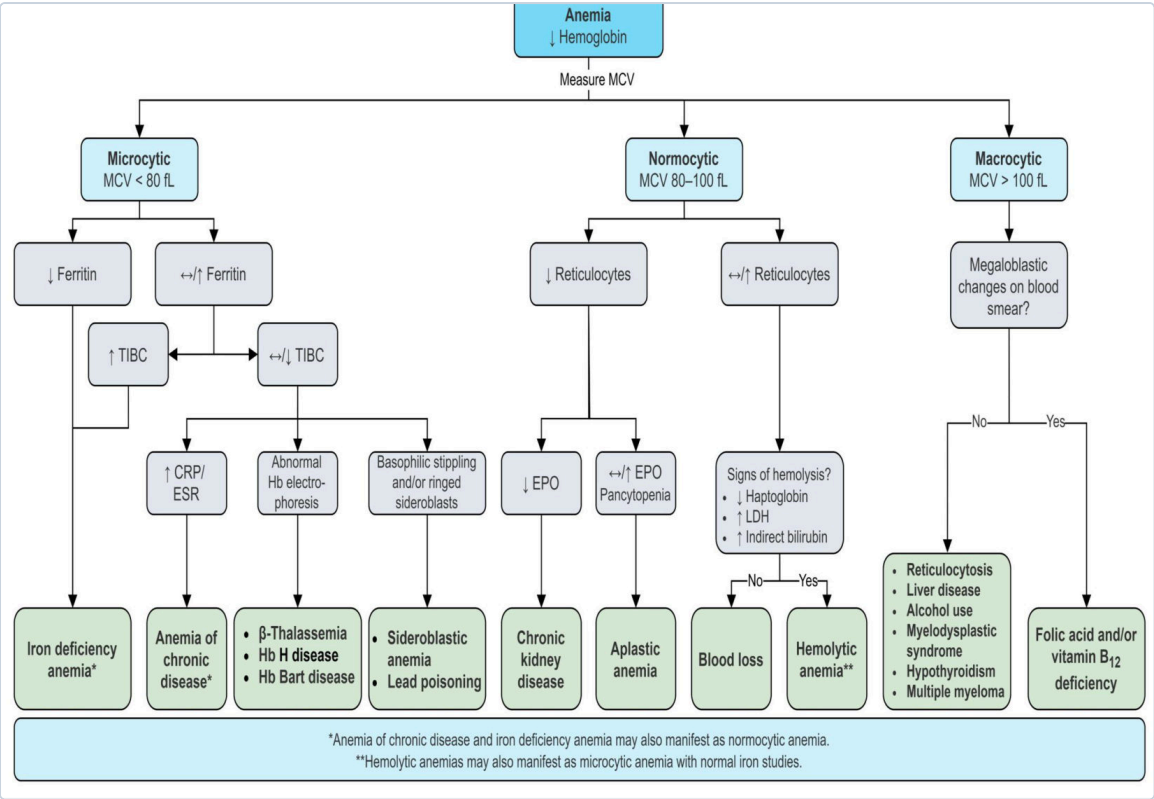
1. **Treat the cause**
2. ⭐ **Transfuse if: Hb < 7 g/dL**, **or HF**, **or ischaemic heart disease**

Classification (slide 27):

- **By aetiology:** deficiency · marrow failure · haemolytic · haemorrhagic
- **By red-cell type:**

Normocytic normochromic	Microcytic hypochromic	Macrocytic normochromic
Marrow failure	⭐ **Iron deficiency**	**Megaloblastic**: **B12 or folate**
Acute haemorrhage	**Thalassaemia**	**Non-megaloblastic**: **liver failure, hypothyroidism, myelodysplasia, alcoholism**
Haemolysis	**Sideroblastic**	
	Anaemia of chronic disease	
	Chronic lead poisoning	

Mnemonic for microcytic anaemia (slide 42), "IRON LAST": IRON deficiency · **L**ead · **A**naemia of chronic disease · **S**ideroblastic · **T**halassaemia.



Slide 28: MCV-based diagnostic algorithm for anaemia

Slide 28 algorithm:

Branch	Next test → diagnosis
Microcytic (< 80)	**↓ ferritin** → **IDA** · **N/↑ ferritin** → ↑ TIBC → IDA; N/↓ TIBC → **↑ CRP/ESR** (**ACD**), **abnormal Hb electrophoresis** (**β-thal, HbH, Hb Bart**), **basophilic stippling / ringed sideroblasts** (**sideroblastic, lead**)
Normocytic (80–100)	**↓ retics** → ↓ EPO (**CKD**) or N/↑ EPO + pancytopenia (**aplastic**) · **N/↑ retics** → haemolysis signs (↓ haptoglobin, ↑ LDH, ↑ indirect bilirubin)? Yes = **haemolytic**; No = **blood loss**
Macrocytic (> 100)	**Megaloblastic changes on film?** Yes → **folate/B12** · No → **reticulocytosis, liver disease, alcohol, MDS, hypothyroidism, multiple myeloma**

Footnotes on the slide:

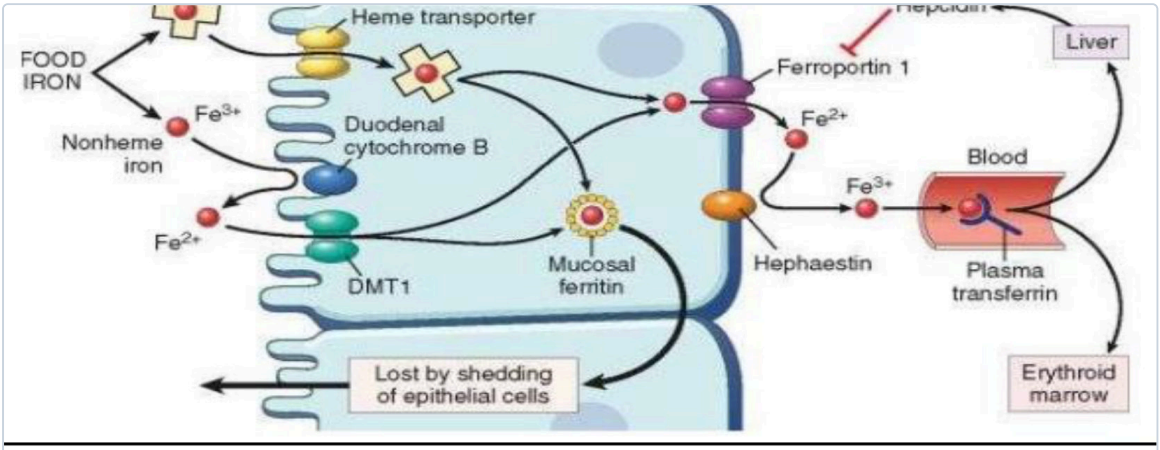
- **ACD and IDA can be normocytic** early on
- **haemolytic anaemias** may appear microcytic with normal iron studies

> 🔁 **Recap: Anaemia general.**
> - **Definition:** Hb < 13 ♂ / < 12 ♀.
> - **Compensation:** hyperdynamic circulation, EPO, **2,3-DPG right shift**, redistribution.
> - **Classification:** by MCV; **transfuse < 7** (or with HF/IHD).
>
> 🔗 **Link:** the commonest microcytic anaemia, and the commonest anaemia worldwide, is iron deficiency.
>
> ▶️ **Up next: Iron deficiency anaemia.**

2.8 Iron Physiology (Slides 30–31) ⭐⭐

Item	Detail
Sources	Meat, liver, spinach

Item	Detail
Requirement	**10 mg/day ingested, ~1 mg absorbed**
Absorption	**Ferric (Fe^{3+}) → ferrous (Fe^{2+})** in the stomach by **HCl**; absorbed in the **duodenum** via the **apoferritin transport system**, which is saturable (**~10 % of intake absorbed**; **mucosal block theory**)
Functions	**Myoglobin & Hb**; haem enzymes (**cytochrome oxidase, catalase**)
Storage	**~1 g** as **ferritin or haemosiderin** in the RES
Circulating ferritin	♀ **10–200 ng/mL**, ♂ **15–400**; **1 µg/L ferritin ≈ 10 mg stored iron**
Transport	Bound to **transferrin**, normally **~30 % saturated**



- ***Transport system:*** iron is transported in the blood bound to transferrin, which is only 30% saturated by iron. Unbound transferrin is called total iron binding capacity (TIBC). Normally transferrin carries 100 Ug/ml of iron & binding capacity is 300 Ug/ml (TIBC = 70%.)

Slide 31: Iron absorption and recycling (DMT1, ferroportin, hepcidin, transferrin)

[added] **Slide 31 diagram mechanics:**

- **DMT1** takes Fe^{2+} into the enterocyte
- iron leaves via **ferroportin**, then is oxidised by **hephaestin** and loaded onto transferrin
- ⭐ **Hepcidin** (from the liver) **blocks ferroportin** → iron trapped in enterocytes and macrophages. This is the mechanism of **anaemia of chronic disease**.

> ⚠️ **Slide 31 error:** it says "unbound transferrin is called TIBC … TIBC = 70 %".
> - **TIBC is the total binding capacity**: **~300 µg/dL** (slide: "300", units given as µg/mL).
> - The **unsaturated** fraction (~70 %) is the **UIBC**.
> - **Serum iron ~100 µg/dL ÷ TIBC 300 = ~30 % saturation.**
> - Slide 34 repeats the error ("iron binding capacity N = 70 %").

2.9 IDA: Aetiology (Slide 32) ⭐

Mechanism	Causes
↓ Intake	Starvation, dysphagia, prolonged parenteral feeding without iron; **infants on milk only after 6 months** (maternal stores used up)
↓ Absorption	**Gastrectomy**, gastric carcinoma, **atrophic gastritis**, **malabsorption** (*[added]* **coeliac**), excess **phytate/phosphate** *[added: PPIs, tea]*
⭐ **↑ Loss**	**Chronic bleeding**: peptic ulcer, varices, gastritis, **colon cancer**, piles, haematuria, haemoptysis, **menorrhagia**, metrorrhagia; ⭐ ***Ancylostoma duodenale*** (hookworm; "common cause in Egypt")
↑ Requirement	**Pregnancy, lactation**, puberty, convalescence

2.10 IDA: Clinical Picture (Slide 33) ⭐

I. General anaemia features.

II. Specific features. The enzyme defect hits **high-turnover epithelial cells** first:

- **red glazed tongue** (atrophic glossitis) *[added: angular stomatitis]*
- dyspepsia
- ⭐ **koilonychia** (spoon nails) with loss of lustre
- **hair loss**
- ⭐ **pica** (eating unusual substances; *[added]* **pagophagia** = ice)
- *[added]* **Plummer–Vinson syndrome** = IDA + **oesophageal web** (dysphagia) + glossitis; premalignant (post-cricoid SCC)

III. Features of the cause, e.g. **hookworm**:

- epigastric pain, constipation
- **parotid enlargement** ("endemic parotitis")
- perverted appetite
- **pellagra** (co-existing malnutrition)

2.11 IDA: Investigations (Slide 34) ⭐⭐

Steps:

1. **↓ Hb, ↓ Hct, ↓ RBCs, ↑ ESR**
2. **Type: microcytic hypochromic**: ↓ MCV, ↓ MCHC
3. **Cause (iron studies):**

Test	IDA	Normal (slide)
Serum iron	↓	**70–140 µg/dL**
TIBC	⭐ ↑	(slide: "N = 70 %"; see the error flagged above)
Transferrin saturation	↓	**25–50 %**
⭐ **Ferritin**	↓ (most specific)	
Bone marrow	**Hypercellular with normoblasts** *[added: ⭐ absent stainable iron (Prussian blue) = gold standard]*	

4. **Aetiology:**

- hookworm → **eosinophilia**, stool for worm/ova
- **faecal occult blood**
- **endoscopy** (*[added]* bidirectional: OGD + colonoscopy in men and post-menopausal women)
- **platelets may rise** with active blood loss

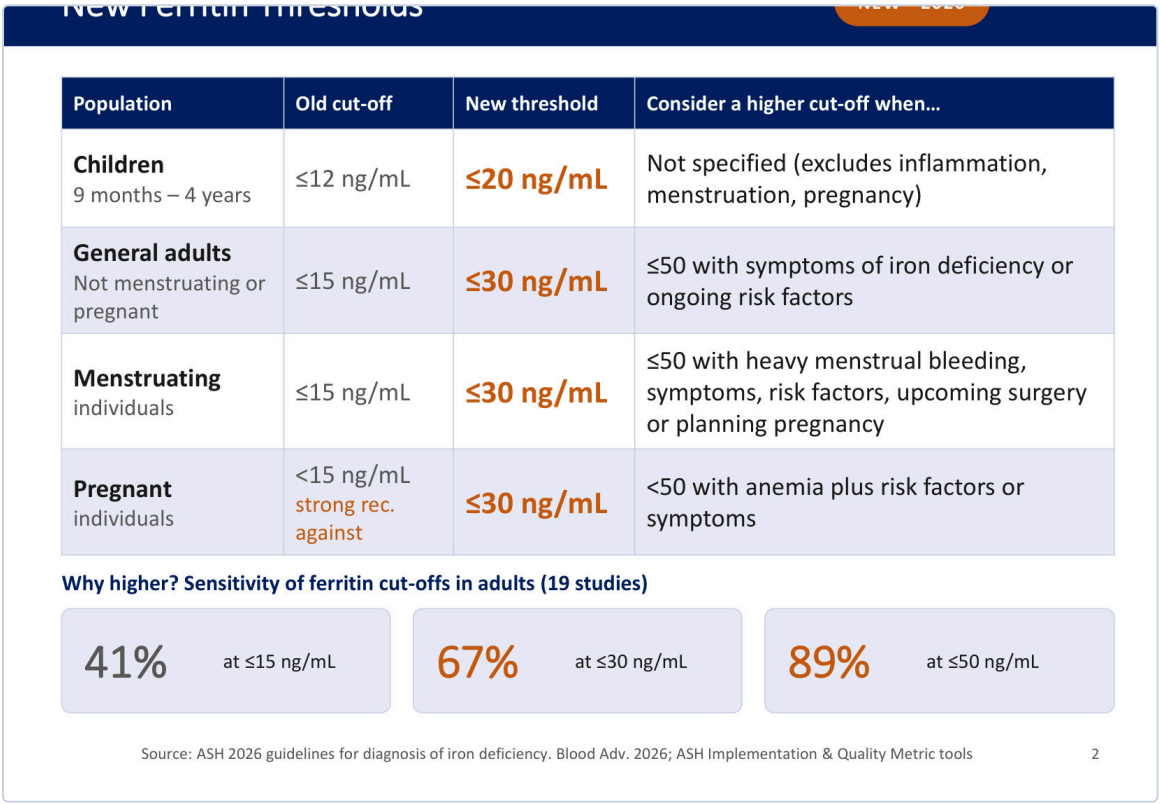
5. ⭐ **Therapeutic test:** iron → **reticulocytosis**

2.12 New ASH 2026 Ferritin Guidelines (Slides 35–37) ⭐⭐

> *[added]* I can't verify this guideline independently; it postdates my reliable knowledge. It is reproduced exactly as on the slides. The lecturer clearly considers it high-yield ("NEW").

What changed (slide 35):

1. **Ferritin cut-offs go up in every group studied**
2. **With inflammation, use ferritin + TSAT together**
3. **Labs should use clinical decision limits**, not population reference ranges

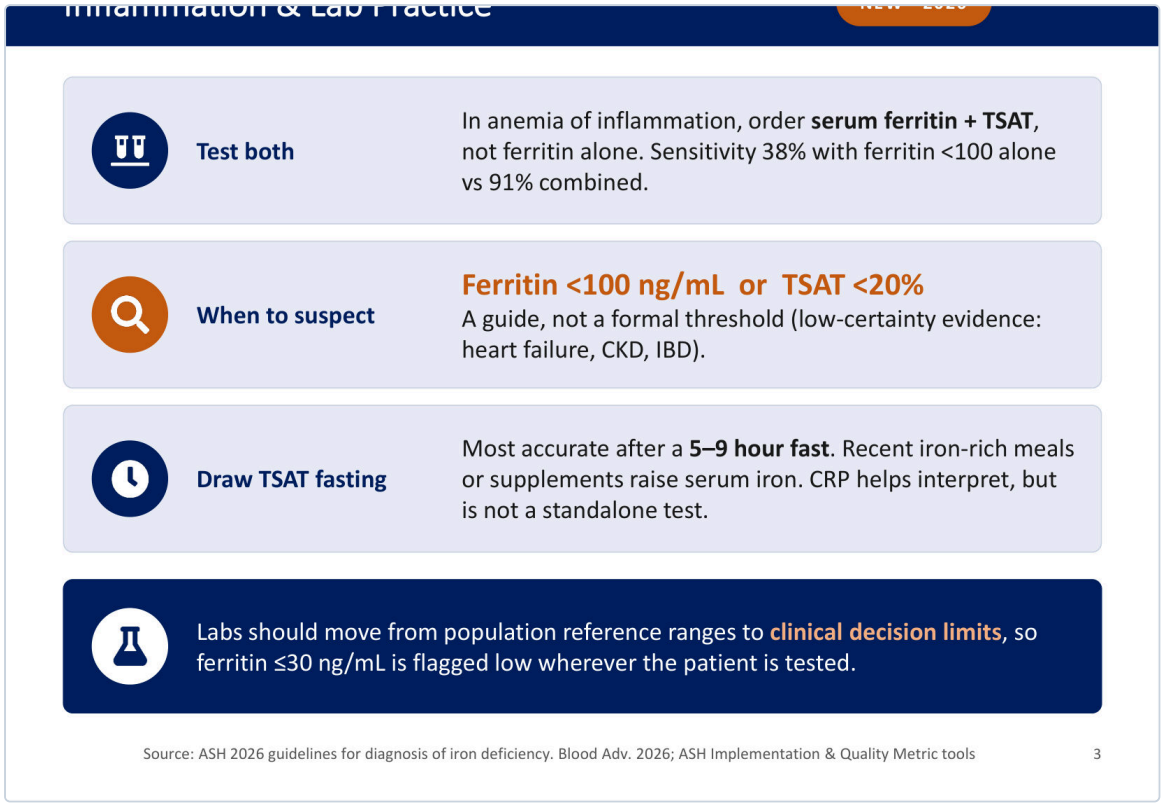


Slide 36: New ferritin thresholds by population

Population	Old cut-off	⭐ New threshold	Consider a higher cut-off when…
Children 9 mo–4 y	≤ 12 ng/mL	**≤ 20 ng/mL**	Not specified (excludes inflammation, menstruation, pregnancy)
General adults (not menstruating/pregnant)	≤ 15	**≤ 30**	**≤ 50** with symptoms of ID or ongoing risk factors
Menstruating	≤ 15	**≤ 30**	**≤ 50** with heavy menstrual bleeding, symptoms, risk factors, upcoming surgery, planning pregnancy
Pregnant	< 15 (strong rec. against)	**≤ 30**	**< 50** with anaemia plus risk factors or symptoms

Why higher? Sensitivity of ferritin cut-offs in adults (19 studies):

- **41 %** at ≤ 15
- **67 %** at ≤ 30
- **89 %** at ≤ 50



Slide 37: Inflammation and lab practice

Inflammation & lab practice (slide 37):

- ⭐ **Test both** ferritin and TSAT in anaemia of inflammation, **not ferritin alone**. Sensitivity is **38 % with ferritin < 100 alone vs 91 % combined**.
- **When to suspect ID with inflammation: ferritin < 100 ng/mL or TSAT < 20 %**. This is a guide, not a formal threshold (low-certainty evidence: **HF, CKD, IBD**).
- **Draw TSAT fasting** (most accurate after a **5–9 h fast**); recent iron-rich meals or supplements raise serum iron. **CRP helps interpretation but is not a standalone test.**
- Labs should flag **ferritin ≤ 30 ng/mL as low** wherever the patient is tested.

2.13 IDA: Treatment (Slides 38–39) ⭐⭐

I. Iron replacement

1. Oral iron (**ferrous gluconate, ferrous sulphate**):

- **Side effects:** ⭐ **black stool** (DD melaena), **constipation**, nausea, vomiting, gastric ulcers
- **Duration: 4–10 weeks** until Hb is normal, **then small doses for 3–6 months to refill stores**
- **Expected response:**
  - ⭐ **reticulocytes rise on day 4**, peak/persist ~**12 days**
  - ⭐ **Hb starts rising after 1 week**, by **~1 g/dL per week**
- *[added]* Alternate-day dosing improves absorption (less hepcidin rise). Take with vitamin C; avoid tea and PPIs.

2. Parenteral iron (**iron dextran IM/IV, iron sorbitol IM**; *[added: modern IV = ferric carboxymaltose, iron sucrose]*):

- **Indications** (oral not usable):
  - **peptic ulcer**
  - **malabsorption**
  - **oral side effects**
  - ⭐ **need for rapid correction (late pregnancy)**
  - **failure of oral iron**
- **Complications: allergy/anaphylaxis** (esp. iron dextran), thrombophlebitis, **haemochromatosis** (iron overload)

II. Treat the cause:

- hookworm → **mebendazole 100 mg BID × 3 days**
- treat peptic ulcer

III. Transfusion if:

- **Hb < 7 g/dL**
- ⭐ **IHD or HF with Hb < 10 g/dL**

> 🔁 **Recap: IDA.**
> - **Absorption:** 10 mg in, 1 mg absorbed (duodenum, Fe^{2+}, needs HCl).
> - **Causes:** chronic blood loss (GI, menorrhagia, hookworm), ↓ absorption, ↑ need.
> - **Clinical:** koilonychia, pica, glossitis.
> - **Labs:** ↓ iron, ↑ **TIBC**, ↓ TSAT, ↓ **ferritin** (ASH 2026: **≤ 30**; ferritin + TSAT if inflamed).
> - **Treatment:** oral iron 4–10 weeks + 3–6 months (retics day 4, Hb +1 g/week); IV when oral fails or speed is needed; transfuse < 7 (< 10 with IHD/HF).
>
> 🔗 **Link:** the other microcytic anaemias look like IDA clinically but differ in **iron studies**.
>
> ▶️ **Up next: Sideroblastic anaemia, ACD and the microcytic DD.**

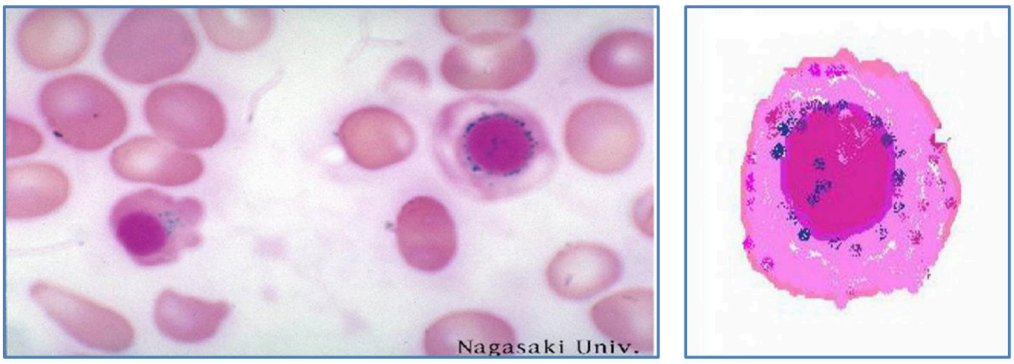
2.14 Sideroblastic Anaemia (Slide 40) ⭐

Pathophysiology:

- iron **cannot be incorporated into haem** because of a defect in **ALA synthase** (*[added]* the rate-limiting haem enzyme, B6-dependent)
- unused iron deposits as **haemosiderin in mitochondria** → ⭐ **ringed sideroblasts** (iron granules ringing the nucleus)
- most are destroyed in the marrow; some reach the blood

Pathophysiology:

- Failure of incorporation of iron in heme synthesis due to defect in gamma amino-levulinic acid synthetase (ALA).
- The utilized iron will be deposited as hemsiderin in the mitochondria with formation of ringed sideroblasts. These cells are destroyed in the BM but some may appear in the peripheral blood.



Slide 40: Pappenheimer bodies and a ringed sideroblast (Prussian blue)

> ⚠️ **Slide error:** "**gamma** amino-levulinic acid synthetase" should be **δ-aminolevulinic acid (ALA) synthase**.

[added] **Causes:**

- hereditary: X-linked ALAS2
- acquired: **alcohol, isoniazid (B6 deficiency), lead**, copper deficiency, **MDS (RARS)**
- treat with **pyridoxine** where relevant

2.15 Anaemia of Chronic Disease (Slide 41) ⭐

Item	Detail
Aetiology	**Chronic infection (TB)**, **chronic inflammation (RA)**, **malignancy**
Pathogenesis	↓ RBC survival · **reduced EPO response** · ⭐ **↓ iron release from stores** to erythroblasts (*[added]* **hepcidin ↑**)
Clinical	As IDA
Investigations	As IDA **but**: ⭐ **↓ TIBC**, marrow with **excess stainable iron**, **↑ acute-phase reactants**
Treatment	Of the cause

> ⚠️ Slide 41 writes "excess stainable **sideroblasts**" and "Reduced **EPF**". ACD has **excess stainable iron in macrophages** (not ringed sideroblasts) and a blunted **EPO** response.

2.16 DD of Microcytic Hypochromic Anaemia (Slides 42–43) ⭐⭐⭐

	Iron deficiency	Anemia of chronic disease	Sideroblastic anemia	Thalassemia / Hb-pathies
Mechanism	↓ Iron → ↓ heme	Iron trapped (↑ hepcidin) + ↓ EPO	↓ Porphyrin (heme) synthesis	↓ Globin chain synthesis
RED CELLS				
MCV / MCHC	N → ↓	N → ↓	N → ↓	N → ↓
Retic count	N → ↓	N → ↓	N	↑
Smear	Aniso-/poikilocytosis, target cells	Usually nonspecific	Pappenheimer bodies (ring sideroblasts in marrow)	Many target cells (more than IDA)
IRON STUDIES				
Serum iron	↓	↓	↑	N → ↑
TIBC	↑	N → ↓	N → ↓	N → ↓
% Saturation	↓	↓	↑↑	N
Ferritin	↓	N → ↑	↑	N → ↑
Hepcidin	↓	↑	↓	↓
KEY CLUE	Ferritin ↓ + TIBC ↑ (ASH 2026: ferritin ≤30)	Ferritin N/↑ + TIBC ↓ (hepcidin ↑)	Iron & saturation very ↑, ring sideroblasts	Iron normal, target cells → Hb electrophoresis

Student pearl

The causes of microcytic anemia can be remembered with **IRON LAST**: **IRON** deficiency, Lead poisoning, Anemia of chronic disease, Sideroblastic anemia, Thalassemia.

Slide 42: Microcytic DD table

	Iron deficiency	ACD	Sideroblastic	Thalassaemia / Hb-pathies
Mechanism	↓ iron → ↓ haem	**Iron trapped (↑ hepcidin) + ↓ EPO**	↓ porphyrin (haem) synthesis	↓ globin chain synthesis
MCV / MCHC	N → ↓	N → ↓	N → ↓	N → ↓
Retic count	N → ↓	N → ↓	N	↑
Smear	Aniso/poikilocytosis, target cells	Usually non-specific	**Pappenheimer bodies** (ringed sideroblasts in marrow)	**Many target cells** (more than IDA)
Serum iron	↓	↓	↑	N → ↑
TIBC	⭐ ↑	⭐ **N → ↓**	N → ↓	N → ↓
% saturation	↓	↓	↑↑	N
Ferritin	⭐ ↓	**N → ↑**	↑	N → ↑
Hepcidin	↓	↑	↓	↓
Key clue	**Ferritin ↓ + TIBC ↑** (ASH 2026: ferritin ≤ 30)	**Ferritin N/↑ + TIBC ↓** (hepcidin ↑)	**Iron and saturation very ↑, ringed sideroblasts**	**Iron normal, target cells → Hb electrophoresis**

Clinical pearls (slide 43):

- **IDA and ACD** can be **normocytic early** and **microcytic later**.
- **Marrow failure** (myeloproliferative malignancy, MDS) can be **micro-, normo- or macrocytic**.
- ⭐ **IDA and thalassaemia trait** are the **commonest causes of microcytosis**.

- **Basophilic stippling** → **lead poisoning or sideroblastic anaemia**. **Ringed sideroblasts are usually absent in lead poisoning**, which helps separate them.
- ⚠️ **Caution: low ferritin confirms IDA**, but **high ferritin does not rule it out** (ferritin is an acute-phase reactant).

2.17 Haemorrhagic Anaemia (Slide 44) ⭐

Chronic haemorrhage: behaves as **IDA** (microcytic hypochromic).

Acute haemorrhage (> 1 L lost acutely):

Time	Finding
Immediately	**Proportional loss of RBCs and plasma → PCV NORMAL**; ↓ platelets and neutrophils within hours
Within 1 day	Interstitial fluid moves into vessels → **haemodilution → ↓ PCV**
After 3–4 days	↑ marrow activity → **reticulocytosis**

⭐ **Pearl: Hb and Hct can be normal initially** after acute haemorrhage. They fall only after plasma volume is restored (spontaneously or by IV fluids). *[added]* Shock lecture link: never use the first Hb to judge acute blood loss.

2.18 Case & Summary (Slides 45–47)

See Q1 in §7.

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3. High-Yield Comparison Tables

Table A: Anaemia by MCV

[slides 25, 27, 28]

Microcytic (< 80)	Normocytic (80–100)	Macrocytic (> 100)
IDA, ACD (late), thalassaemia, sideroblastic, lead	Marrow failure, acute haemorrhage, haemolysis, CKD, early IDA/ACD	**Megaloblastic**: B12, folate
		Non-megaloblastic: liver, alcohol, hypothyroidism, MDS, myeloma, reticulocytosis

Table B: Iron Studies Cheat Sheet

See §2.16. Remember: **TIBC moves opposite to ferritin**: IDA = ferritin ↓ / TIBC ↑; ACD = ferritin ↑ / TIBC ↓.

Table C: Oral vs Parenteral Iron

	Oral	Parenteral
Agents	Ferrous sulphate, gluconate	Iron dextran (IM/IV), iron sorbitol (IM)
Use	First line	PU, malabsorption, oral intolerance, **late pregnancy**, oral failure

	Oral	Parenteral
Side effects	**Black stool**, constipation, GI upset, ulcers	**Anaphylaxis**, thrombophlebitis, iron overload
Duration	4–10 weeks + **3–6 months** for stores	Calculated total dose *[added]*

Table D: Reticulocytes

↑ (reticulocytosis)	Absent / low
Haemolysis, acute bleed (after 3–4 days), **response to treatment** (iron day 4)	**Aplastic anaemia**, untreated deficiency, ACD, CKD

Table E: Transfusion Thresholds (slides 26, 39)

Situation	Transfuse at
General	**Hb < 7 g/dL**
IHD or HF	**Hb < 10 g/dL** (slide 39)

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Low ferritin = IDA	**High ferritin does NOT exclude IDA** (acute-phase reactant) → add **TSAT**; with inflammation suspect ID if ferritin < 100 or TSAT < 20 %
IDA is microcytic	**Early IDA/ACD is normocytic**
Microcytic anaemia = iron deficiency	**Thalassaemia trait** is equally common → **normal ferritin, normal RDW, target cells** → Hb electrophoresis
Anaemia raises ESR	**Sickle cell and HS lower ESR**
Acute bleed → low Hb	**Hb/Hct normal at first**; falls within 24 h
Basophilic stippling	**Lead** (no ringed sideroblasts) vs **sideroblastic** (ringed sideroblasts) vs **thalassaemia**
Oral iron → black stool	**Not melaena**: stool is formed and not tarry *[added]*; check FOBT if in doubt
Transfuse at Hb < 7	**IHD/HF → < 10**
Iron absorption	Needs **acid** → gastrectomy, atrophic gastritis, *[added]* PPIs ↓ absorption
Ferritin cut-off ≤ 15 (old)	**ASH 2026: ≤ 30** (adults, menstruating, pregnant); **≤ 20** in children 9 mo–4 y

Pattern recognition:

- **elderly man + IDA** → GI malignancy work-up (OGD + colonoscopy) *[added]*
- **NSAID use + IDA** → peptic ulcer/gastritis bleeding
- **koilonychia + dysphagia + glossitis** → Plummer–Vinson *[added]*
- **microcytosis + very high iron + ringed sideroblasts** → sideroblastic
- **microcytosis + ferritin ↑ + TIBC ↓ + RA/TB** → ACD

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5. Clinical Correlations

- **Elderly man with IDA** (the slide case): **diet + NSAIDs** explain it, but an *[added]* **endoscopic search for GI malignancy** is still mandatory.

- **Black stool on oral iron vs melaena:** ask about tarry, sticky, foul stool and haemodynamic change. *[added]* Iron stool is dark green-black and formed.
- **IHD/HF patients** tolerate anaemia badly (angina, anaemic HF) → transfuse at **Hb < 10**.
- **Pregnancy:**
  - iron needs rise
  - **IV iron in late pregnancy** when rapid correction is needed
  - ASH 2026 ferritin threshold **≤ 30**, or **< 50** with anaemia + risk factors/symptoms
- **Hookworm** (*Ancylostoma duodenale*):
  - eosinophilia + IDA
  - treat with **mebendazole 100 mg BID × 3 days** + iron
  - travel/endemic exposure (bridge to Travel Medicine)
- **Lab practice:**
  - draw **TSAT fasting**
  - **CRP** helps interpret ferritin
  - ferritin **≤ 30 flagged low** wherever tested

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6. Rapid Revision (Last-Day)

- **Plasma & serum:** plasma 90 % water, 5–7 % protein (albumin, globulin, fibrinogen). **Serum = plasma − clotting factors.**
- **Haematopoiesis sites: yolk sac (wk 3) → liver/spleen (mo 3) → marrow (from mo 4)**. Adult marrow = ends of long bones + flat bones. Extramedullary = liver/spleen under stress.
- **Erythroid maturation:** proerythroblast → erythroblast → normoblast → **reticulocyte (RNA)** → RBC.
- **Requirements: iron, B12, folate, amino acids, EPO**.
- **RBC normals:**
  - count ♂ 4.5–6, ♀ 3.8–4.8 × 10^6
  - diameter **7.2 µm** (< 6.7 micro, > 8 macro, > 12 megalocyte)
  - Hct **45 %**
  - MCV **78–98** (use < 80 / > 100)
  - Hb ♂ 13.5–16, ♀ 11.5–13.5
  - MCHC **32–36 %**
  - MCH **~30 pg**
  - retics **0.2–2 %**
- **ESR:**
  - normal ♂ 5, ♀ 10 mm in the 1st h (Westergren)
  - ↑ fibrinogen/globulin, ↓ albumin
  - ↓ in polycythaemia, afibrinogenaemia, HF, **sickle cell, HS**
  - **> 100: malignancy, collagen, TB**
  - monitoring, not diagnosis
- **Blood film:**
  - **stippling: lead, thal**
  - **Howell–Jolly: megaloblastic, asplenia**
  - **↑ RDW: IDA**
- **Anaemia definition: Hb < 13 ♂ / < 12 ♀**.
- **Compensation:** hyperdynamic circulation, EPO, **2,3-DPG → right shift**, redistribution.
- **General anaemia features:** pallor, tinnitus, palpitations, **water-hammer pulse, flow murmur, anaemic HF**, oedema.

- **Transfusion: < 7**, or **< 10 with IHD/HF**.
- **Microcytic causes = IRON LAST: IDA, lead, ACD, sideroblastic, thalassaemia**.
- **Macrocytic causes:** megaloblastic (B12/folate) vs non-megaloblastic (liver, alcohol, hypothyroid, MDS).
- **Iron physiology:**
  - 10 mg/day in, **1 mg absorbed**
  - **Fe^{3+} → Fe^{2+} by HCl**, duodenum
  - 1 g stored (ferritin/haemosiderin)
  - transferrin ~30 % saturated
  - **hepcidin blocks ferroportin**
- **IDA causes:** chronic blood loss (PU, colon cancer, menorrhagia, **hookworm**), gastrectomy/atrophic gastritis/malabsorption, ↑ need (pregnancy, puberty), milk-only infants > 6 months.
- **IDA features: glossitis, koilonychia, pica**, hair loss. Hookworm: parotid enlargement, pellagra.
- **IDA labs:**
  - **↓ iron (N 70–140), ↑ TIBC, ↓ TSAT (N 25–50 %), ↓ ferritin**
  - marrow: no stainable iron *[added]*
  - **therapeutic test → retics**
- **ASH 2026 ferritin thresholds:**
  - **≤ 30** (adults, menstruating, pregnant); **≤ 20** children 9 mo–4 y
  - **≤ 50** if symptoms/risk
  - inflammation: **ferritin + TSAT** (< 100 or < 20 %)
  - sensitivity 41/67/89 % at 15/30/50
- **IDA treatment:**
  - **oral ferrous sulphate/gluconate** for 4–10 weeks + 3–6 months
  - **retics day 4**, Hb +1 g/dL/week from week 1
  - black stool, constipation
  - **IV iron**: PU, malabsorption, intolerance, **late pregnancy**, failure; risk of anaphylaxis
  - hookworm: **mebendazole 100 mg BID × 3 d**
- **Sideroblastic anaemia: δ-ALA synthase** defect → **ringed sideroblasts**, **↑ iron, ↑↑ TSAT, ↑ ferritin**.
- **ACD:** TB/RA/cancer; **hepcidin ↑**; **ferritin N/↑, TIBC ↓**, marrow iron ↑.
- **Thalassaemia:** iron normal, **target cells**, ↑ retics → **Hb electrophoresis**.
- **Acute haemorrhage: PCV normal at first** → ↓ in 24 h (haemodilution) → retics at **3–4 days**.

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7. Exam-Style Questions

Q1. (Lecture case) A 78-year-old widower has 6 months of fatigue, headache and occasional upper abdominal pain. He takes HCTZ, metformin and **ibuprofen** for shoulder pain, and his diet is **mostly potatoes and eggs**. He has pallor and a mild postural tremor; rectal exam is normal. The film shows RBCs with **increased central pallor and anisocytosis**. Underlying cause? a) Lead poisoning · b) Folate deficiency · c) Marrow failure · d) Intravascular haemolysis · e) ↓ EPO · f) B12 deficiency · **g) Iron deficiency**

> **Answer: g** *[reasoning added; key not on slides]*.
>
> - **Increased central pallor = hypochromia**; anisocytosis = ↑ RDW. Together they mean **IDA**.
> - Mechanism: **poor intake** (potatoes/eggs are poor in haem iron) plus **NSAID-induced GI blood loss** (epigastric pain).
> - **Why not the others:**
> - **Lead (a):** abdominal pain and tremor tempt, but there is no exposure history, and you'd expect **basophilic stippling**.
> - **Folate/B12 (b, f):** macrocytosis and hypersegmented neutrophils, not hypochromia.
> - **(c)** marrow failure → pancytopenia. **(d)** intravascular haemolysis → jaundice, haemoglobinuria. **(e)** ↓ EPO (CKD) → normocytic.
> - *[added]* An older man with IDA still needs **OGD + colonoscopy**.

Q2. A woman with RA has Hb 10 g/dL, MCV 76 fL, serum iron ↓, **TIBC ↓**, **ferritin 180 ng/mL**, CRP ↑. Diagnosis? A. IDA · **B. Anaemia of chronic disease** · C. β-thalassaemia trait · D. Sideroblastic anaemia · E. Lead poisoning

> **Answer: B** (slides 41–42).
>
> - Ferritin N/↑ + **TIBC ↓** + inflammation, driven by **hepcidin**.
> - **Caveat** (slide 37): with inflammation, coexisting ID is suspected if **ferritin < 100 or TSAT < 20 %**. Here ferritin is 180, so check TSAT.

Q3. A patient starts oral ferrous sulphate for IDA. When should the **reticulocyte count** first rise, and how fast should Hb climb? A. Day 1; 3 g/dL/week · **B. Day 4; ~1 g/dL/week from week 1** · C. Week 3; 0.5 g/dL/month · D. Day 10; 2 g/dL/day · E. No change for 3 months

> **Answer: B** (slide 38).
>
> - Retics rise on day **4** and persist ~12 days. Hb starts rising after **1 week**, by **~1 g/dL/week**.
> - Continue for **3–6 months after Hb normalises** to refill stores.

Q4. Microcytic anaemia with **serum iron ↑, TSAT ↑↑, ferritin ↑** and **ringed sideroblasts** in the marrow. Defect? A. Globin chain synthesis · **B. δ-aminolevulinic acid synthase (haem synthesis)** · C. Hepcidin excess · D. Ferroportin loss · E. Spectrin

> **Answer: B** (slides 40, 42).
>
> - Iron can't enter haem → mitochondrial iron ring. *[added]* Think alcohol, isoniazid, lead, MDS; try pyridoxine.

Q5. A 60-year-old with known IHD has Hb 8.5 g/dL from chronic GI loss, with exertional angina. Most appropriate? A. Oral iron only and recheck in 3 months · **B. Transfuse (Hb < 10 with IHD), then iron + find the source** · C. Wait until Hb < 7 · D. IV iron dextran without investigation · E. EPO

> **Answer: B** (slide 39). Transfusion threshold is **< 10 g/dL in IHD/HF**, versus < 7 in general.

Q6. A non-pregnant, non-menstruating adult has fatigue, Hb normal, ferritin **24 ng/mL**, CRP normal. According to the ASH 2026 guidelines on the slides: A. Normal; ferritin > 15 · **B. Iron deficient; the new adult threshold is ≤ 30 ng/mL** · C. Needs bone-marrow iron stain · D. Iron deficient only if TSAT < 5 % · E. ACD

> **Answer: B** (slides 35–36). The old threshold was ≤ 15. ≤ 50 may be used with symptoms or ongoing risk.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
"TIBC = 70 %" (slides 31, 34)	❌ TIBC is a **capacity (~300 µg/dL)**; **saturation ~30 %**; the 70 % is the **unsaturated** fraction
Normal ferritin excludes IDA	❌ Ferritin is an **acute-phase reactant**; use **ferritin + TSAT** in inflammation
Ferritin ≤ 15 = cut-off	⚠️ **ASH 2026 (slides): ≤ 30** in adults, menstruating and pregnant
ACD has high TIBC	❌ **ACD: TIBC ↓**; IDA: TIBC ↑
All microcytic anaemia → iron	❌ **Thalassaemia trait** (normal ferritin, target cells) gets **no iron**; check Hb electrophoresis
Acute GI bleed with normal Hb → minor	❌ **Hb/Hct normal initially**; falls after haemodilution
ESR diagnoses disease	❌ It **monitors** disease and treatment
Anaemia always ↑ ESR	❌ **Sickle cell, HS → ↓ ESR**
Stop oral iron when Hb normalises	❌ Continue **3–6 months** to refill stores
Black stool on iron = melaena	⚠️ Expected side effect; melaena is tarry and with haemodynamic change *[added]*
Transfuse everyone at Hb < 10	❌ **< 7** generally; **< 10 only with IHD/HF**
Ringed sideroblasts in lead poisoning	❌ Usually **absent** in lead; present in sideroblastic anaemia
"Gamma"-ALA synthase (slide 40)	❌ **δ-ALA synthase**
MCV < 78 vs < 80	⚠️ The slides use both; **< 80** is standard
ACD marrow has "excess sideroblasts" (slide 41)	❌ **Excess stainable iron in macrophages**; ringed sideroblasts = sideroblastic anaemia